Aung

Ankle (2-DOF) & Knee (1-DOF) — Peak Torque Calculations

Figure 4.1 body proportions | H = 1.6 m | m = 50.0 kg

1 Body Segment Dimensions

Heights = joint centres measured from the ground

L_foot = 0.152 x H = 0.152 x 1.6 = 0.2432 m = 243.2 mm

W_foot = 0.055 x H = 0.055 x 1.6 = 0.0880 m = 88.0 mm

h_ankle = 0.039 x H = 0.039 x 1.6 = 0.0624 m = 62.4 mm

h_knee = 0.285 x H = 0.285 x 1.6 = 0.4560 m = 456.0 mm

h_hip = 0.530 x H = 0.530 x 1.6 = 0.8480 m = 848.0 mm

L_shank = (0.285-0.039) x H = 0.246 x 1.6 = 0.3936 m = 393.6 mm

L_thigh = (0.530-0.285) x H = 0.245 x 1.6 = 0.3920 m = 392.0 mm

2 Ankle Joint Position & Moment Arms

Ankle sits at 37% of foot length from heel (human body proportion)

x_ankle = 0.37 x L_foot = 0.37 x 0.2432 = 0.0900 m = 90.0 mm (from heel)

r_FFE = L_foot - x_ankle = 0.2432 - 0.0900 = 0.1532 m = 153.2 mm

(push-off moment arm — ankle joint to toe contact)

r_FAA = W_foot / 2 = 0.0880 / 2 = 0.0440 m = 44.0 mm

(side-tilt moment arm — ankle joint to foot edge)

3 Ground Reaction Force

BW = m x g = 50.0 x 9.81 = 490.50 N

F_static = 1.0 x BW = 490.50 N (standing still)

F_walking = 1.5 x BW = 735.75 N <- design basis

F_landing = 2.5 x BW = 1226.25 N (stair landing)

F_trip = 3.5 x BW = 1716.75 N (trip / fall)

Note: 1.5x is real physics not a safety factor

4 Ankle FFE Torque (Front-Back Walking Motion)

At push-off only toe touches. GRF acts upward at toe.

Moment arm = horizontal distance from ankle joint to toe.

tau_FFE = F_GRF x r_FFE

Quasi-static (1-leg stand):

tau = F_s x x_ankle = 490.50 x 0.0900 = 44.14 N.m

Walking <- design basis:

tau = F_w x r_FFE = 735.75 x 0.1532 = 112.72 N.m

Stair landing:

tau = F_l x r_FFE = 1226.25 x 0.1532 = 187.86 N.m

Trip / impact:

tau = F_t x r_FFE = 1716.75 x 0.1532 = 263.01 N.m

> Ankle FFE peak required: 112.72 N.m (walking governs)

5 Ankle FAA Torque (Side-to-Side Balance)

GRF acts at outer foot edge. Moment arm = half foot width.

tau_FAA = F_walking x (W_foot / 2)

= 735.75 x 0.0440

= 32.37 N.m

Cross-check (40% rule): 0.40 x 112.72 = 45.09 N.m (close match)

> Ankle FAA peak required: 32.37 N.m (walking governs)

6 Differential Coupling (2 motors, both DOF)

Both motors same direction -> FFE (torques add)

Motors opposite directions -> FAA (one motor alone)

FFE combined: each motor >= 112.72 / 2 = 56.36 N.m

FAA single: each motor >= 32.37 N.m

FAA (32.37) > FFE half (56.36) -> FAA governs

> Each ankle motor must peak >= 32.37 N.m

7 Knee KFE Moment Arm (Geometry)

When knee bends by angle theta, it shifts horizontally from GRF line.

$d_{knee} = L_{shank} \times \sin(\theta)$

$L_{shank} = 0.3936\ m = 393.6\ mm$ (from Figure 4.1)

Walking (theta = 15 deg):

$\sin(15\ deg) = 0.2588$

$d_{knee} = 0.3936 \times 0.2588 = 0.1019\ m = 101.9\ mm$

Stair climbing (half-thigh method):

CoM sits roughly above half the thigh when knee is deeply bent.

$d_{knee_stair} = L_{thigh} / 2 = 0.3920 / 2 = 0.1960\ m = 196.0\ mm$

8 Knee KFE Torque

$\tau_{knee} = F_{walking} \times d_{knee}$

Walking (theta = 15 deg):

$\tau_{knee} = 735.75 \times 0.1019 = 74.97\ N.m$

Stair climbing:

$\tau_{stair} = 735.75 \times 0.1960 = 144.21\ N.m$

Knee peak required: 74.97 N.m (walking) | 144.21 N.m (stairs)

9 Knee Angle Effect on Torque

The deeper the bend, the longer the moment arm, the higher the torque.

theta = 0 deg : d = 0.0 mm -> tau = 0.0 N.m (mid-stance — zero)

theta = 15 deg : d = 101.9 mm -> tau = 75.0 N.m (normal walking)

theta = 30 deg : d = 196.8 mm -> tau = 144.8 N.m (fast walking)

theta = 45 deg : d = 278.3 mm -> tau = 204.8 N.m (steep stair)

theta = 90 deg : d = 393.6 mm -> tau = 289.6 N.m (deep squat — max)

10 Summary — All Results

Joint DOF Moment Arm Peak Torque Case

Ankle FFE 153.2 mm 112.72 N.m Walking push-off

Ankle FAA 44.0 mm 32.37 N.m Lateral balance

Knee KFE 101.9 mm 74.97 N.m Walking 15 deg

Knee KFE 196.0 mm 144.21 N.m Stair climbing

11 Actuator Recommendations

Ankle — Two RobStride RS03 (60 N.m each) differential

FFE combined: 60 + 60 = 120 N.m required: 112.72 N.m

SF_FFE = 120 / 112.72 = 1.06 x

FAA single: 60 N.m required: 32.37 N.m

SF_FAA = 60 / 32.37 = 1.85 x

Weight per ankle: 2 x 880 g = 1,760 g Both ankles: 3,520 g

Knee — One RobStride RS04 (120 N.m) per knee

Walking: 120 N.m required: 74.97 N.m

SF_walk = 120 / 74.97 = 1.60 x

Stairs: 120 N.m required: 144.21 N.m

SF_stair = 120 / 144.21 = 0.83 x (gentle stairs only)

Weight per knee: 1,420 g Both knees: 2,840 g

Note: CubeMars AK80-64 (64:1 ratio) NOT backdrivable — avoid for leg joints.

12 Complete Actuator Bill (Per Full Robot)

Ankle L+R: 4 x RobStride RS03 4 x 880 g = 3,520 g = 3.52 kg

Knee L+R: 2 x RobStride RS04 2 x 1,420 g = 2,840 g = 2.84 kg

Total (6 actuators): 6.36 kg